INTUITION_OS_MASTER_SPEC.md

PART 9

CONTEXT_ASSEMBLY_SPEC.md

LEARNING MEMORY · CONTEXT RETRIEVAL · PROMPT CONSTRUCTION

Version 1.0

PURPOSE

This document defines the Context Assembly Engine.

The Context Assembly Engine is responsible for constructing the context sent to every AI prompt.

This is arguably the most important system in Intuition OS.

Bad context produces bad mentorship.

Good context produces personalization.

The mentor itself is not the moat.

The context supplied to the mentor is the moat.

CORE PRINCIPLE

The AI never reads the database directly.

The AI receives a curated context package.

WHY THIS EXISTS

Most AI systems fail because they provide:

Too little context.

or

Too much context.

Both are harmful.

BAD EXAMPLE

```text id="bad1" User Profile

Entire Session History

Entire Evidence History

All Interactions

All Reflections

```
Token explosion.

Noise.

Poor answers.

GOOD EXAMPLE

```text id="good1"
Current Problem

Current Session

Relevant Evidence

Relevant Similar Problems

```

```
Minimal Profile Summary
```

Focused.

Useful.

Efficient.

CONTEXT HIERARCHY

Every AI interaction should receive context in this order.

Priority matters.

PRIORITY 1

CURRENT PROBLEM

Highest priority.

Always included.

Data:

```json id="ctx1" { "title": "", "difficulty": "", "tags": [], "url": "" }

```

Reason:

Current problem drives everything.

PRIORITY 2

CURRENT SESSION

Always included.
```
```

```
---

Data:

```json id="ctx2"
{
  "time_spent": 0,
  "submission_count": 0,
  "submission_history": [],
  "mentor_requests": 0,
  "editorial_viewed": false
}
```

Reason:

Current struggle matters more than old history.

PRIORITY 3

CURRENT INTERACTION

Always included.

Data:

```json id="ctx3" { "user_message": "", "current_escalation_level": 0 }

```

Reason:

This is the immediate learning need.

PRIORITY 4

SIMILAR HISTORICAL PROBLEMS

Conditionally included.
```

```
---

Purpose:

Provide personalized memory.

---

# RETRIEVAL RULES

Retrieve maximum:

```text id="ctx4"
Top 3 Similar Sessions
```

Not more.

SIMILARITY CRITERIA

Weight:

```text id="ctx5" Same Topic = 50%

Same Difficulty = 30%

Recency = 20%

```

Example

Current Problem:

Sliding Window

Medium

Retrieve:

Recent Sliding Window Medium Sessions

```

````
# SIMILAR SESSION FORMAT

```json id="ctx6"
[
  {
    "problem": "",
    "result": "",
    "reflection_summary": ""
  }
]
````

PRIORITY 5

RELEVANT EVIDENCE

Conditionally included.

Purpose:

Support personalization.

RETRIEVAL RULE

Do not inject all evidence.

Retrieve only evidence related to:

Current Topic

Current Difficulty

Current Struggle

Example

Current Problem:
````

Dynamic Programming

Retrieve:

```text id="ctx7" DP-related evidence only

```
Not graph evidence.

Not tree evidence.

EVIDENCE RANKING

Priority:

```text id="ctx8"
Recent

Relevant

High Confidence
```

MAXIMUM EVIDENCE

Maximum:

```text id="ctx9" 5 evidence items

```
per prompt.

PRIORITY 6

PROFILE SUMMARY

Lowest priority.

Purpose:
```
```

```
Long-term context.

---

# FORMAT

```json id="ctx10"
{
  "strengths": [],
  "weaknesses": [],
  "trends": []
}
```

LIMIT

Maximum:

```text id="ctx11" 3 strengths

3 weaknesses

3 trends

```

CONTEXT BUDGET

Mentor prompts should follow:

Current Problem

40%

Current Session

25%

```

```
Current Interaction

15%

Historical Sessions

10%

Evidence

5%

Profile

5%
```

This prevents history from overwhelming present context.

SESSION COMPRESSION

Old sessions must never be injected raw.

Instead:

Compress.

Example

Bad:

Entire reflection.

Entire interaction history.

Good:

```text id="ctx12"

Problem: Longest Substring

Outcome: Solved after TLE

Learning: Optimization bottleneck identified.

````

EVIDENCE COMPRESSION

Bad:

Inject raw evidence records.

Good:

```text id="ctx13"

Recent DP Evidence:

Viewed editorial in 4 of last 5 DP problems.

Needed Hint Level 3 in 3 of last 5 DP problems.
````

PROFILE COMPRESSION

Bad:

Full profile.

Good:

```text id="ctx14"

Current Strength:
```

Graph reasoning independence increasing.

Current Weakness:

DP editorial dependency increasing.

````
---

# MENTOR CONTEXT TEMPLATE

Every mentor call receives:

```json id="ctx15"

{
  "problem": {},
  "session": {},
  "interaction": {},
  "similar_sessions": [],
  "evidence": [],
  "profile_summary": {}
}
````

REFLECTION CONTEXT TEMPLATE

Reflections should NOT use profile history.

Input:

```json id="ctx16"

{ "problem": {}, "session": {}, "submissions": [], "mentor_interactions": [] }

```
Reason:

Reflections should describe what happened.

Not who the user is.

```

```
# PROFILE GENERATION CONTEXT

Input:

```json id="ctx17"

{
  "evidence": [],
  "recent_sessions": [],
  "topic_history": []
}
```

No problem-level details.

Only learning-level details.

RECOMMENDATION CONTEXT

Input:

```json id="ctx18"

{ "weaknesses": [], "recent_evidence": [], "recent_topics": [] }

```

CONTEXT INJECTION RULES

Never inject:

- Full editorial
- Full solution
- Full code
- Raw chat history
- Entire profile

Always inject:

Current problem.
```

````
Current session.

Current interaction.

---

# TOKEN MANAGEMENT

Maximum context size:

```text id="ctx19"

4k–6k tokens
````

for mentor interactions.

Target:

```text id="ctx20"

2k–3k tokens

````
average.

RELEVANCE FILTER

Before injection:

Every context item must answer:

```text id="ctx21"

Will this help the mentor answer the current question?
````

If not:

Remove it.

````

STUCK DETECTION CONTEXT

When user appears stuck:

Inject:

```json id="ctx22"

{ "stuck_reason": "", "stuck_duration": "", "submission_pattern": "" }

```

STUCK DETECTION RULES

Potential triggers:

5 minutes inactivity

OR

2 failed submissions

OR

1 TLE

OR

Explicit help request

Never automatically escalate.

Always ask.

Example:

Need a nudge?

[Yes]

[Keep Working]

```
```

````
# FUTURE EVOLUTION

V1:

Rule-based retrieval.

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V2:

Embedding retrieval.

---

V3:

Learning graph retrieval.

---

V4:

Personalized memory ranking.

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# SUCCESS CRITERIA

A mentor should never feel generic.

A mentor should feel like:

```text id="ctx23"

Someone who remembers
what I struggled with last week.
````

If the mentor feels interchangeable with ChatGPT, the Context Assembly Engine has failed.

END OF PART 9
````